Multivariate Analysis

1. What is Multivariate data?
2. Why MVA needed?
3. Techniques.

1) **Multivariate data:-** It refers to data involving more than one variable measured on each observation.

Ex:- Suppose you are analyzing data on students. For each student, you record:

- Marks in Mathematics
- Marks in Science
- Marks in CS.

So this is a multivariate data because (3 Variables) are recorded on each student.

Univariate	Bivariate	Multivariate
One Variable per observation	Two variables per observation	Height, Weight, Age, BP etc.
Height of students	Height & Weight	

Variable → Random Variables

(Mean values, Variances, Covariances, correlations and other statistical measures)

Structure of MVD:- Stored in data matrix where

Rows: observations (students, products)

Columns: Variables (Height, Income, age)

Student	Math	Science	CS
A	85	78	
B	72	55	
C	65	76	
⋮			

Other examples of MVD:- Weather data - Temp, Wind Speed, Rainfall, Air Quality Index

(1) Patient Health Record:- BP, Cholestrol level, Age, Weight, Heart Rate

(2) Smartphone features: Battery capacity, Camera Quality, Screen Size, RAM, Price., etc.

(3) Employee performance in a company: Exp, No of projects completed, Salary, Worker hours per week.

Multivariate Analysis:- Statistical Multivariate Analysis is concerned with data that consists of sets of measurements on a no. of individuals or objects. ~~It's a branch of statistics~~

Multivariate Analysis is a set of techniques used for analysis of data that contain more than one variables. It is important because most real-world problems are influenced by multiple factors. This analysis allows us to understand interactions, patterns & relationships between variables.

↓

Correlation

If we more emphasized on MVD:- Structure of

- Suppose that we have measurements on p variables for each of n experimental units.
- We will use x_{ij} to denote the observed value of the jth variable ($j=1,2,\ldots p$) on the ith unit ($i=1,2,\ldots n$)
- We will typically gather the information into a $(n \times p)$ matrix-

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1j} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2j} & \cdots & x_{2p} \\ \vdots & & & & & \\ x_{n1} & x_{n2} & \cdots & x_{nj} & \cdots & x_{np} \end{bmatrix}_{n \times p}$$

Descriptive Statistics for MVD:-

It refers to the process of summarizing & analyzing data involving more than one variable at a time.

(1) **Mean Vector**:- Represents the avg. of each variables.

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} \quad \text{or} \quad X' = [X_1 \; X_2 \; \cdots \; X_p]$$

$$\bar{X} = \begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \vdots \\ \bar{X}_p \end{bmatrix} \rightarrow \text{mean vector}$$

(2) **Sample covariance matrix** (covariance matrix)

↓

captures the variability of each variables & the covariates (relationships) between them.

Covariances

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix} \rightarrow \text{always symmetric}$$

$$s_{11} = \frac{1}{n-1} \sum (X_1 - \bar{X}_1)^2$$

$$s_{22} = \frac{1}{n-1} \sum (X_2 - \bar{X}_2)^2$$

$$s_{pp} = \frac{1}{n-1} \sum (X_p - \bar{X}_p)^2$$

$$s_{12} = s_{21} = \frac{1}{n-1} \sum \left[(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) \right] \rightarrow \text{Sample Covariance between } X_1 \text{ \& } X_2$$

$$R = \begin{bmatrix} 1 & -0.36 \\ -0.36 & 1 \end{bmatrix}$$

Try that example:-

Multivariate Distributions:- A multivariate distributions describes the underlying random structure of a vector of random variables.

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_p \end{bmatrix}$$

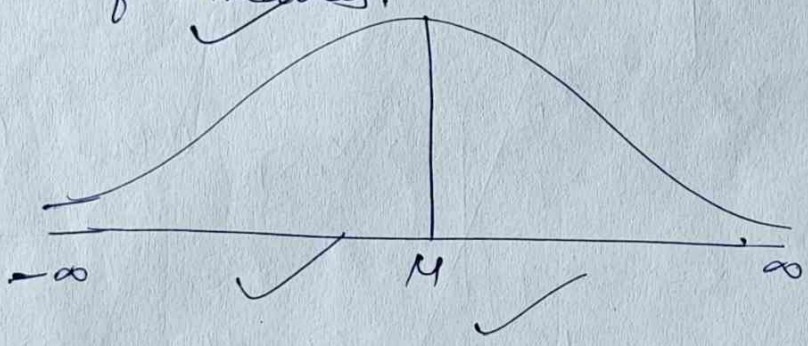
↓ Row vector

Column

$$X_1' = \begin{bmatrix} X_1 & X_2 & \dots & X_p \end{bmatrix}$$

Row vector

- From it we can derive marginal properties of the individual variables.
- It also describes relationships between variables or groups of variables.
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Random Vector & Matrices:- A random vector is a vector whose elements are random variables. ~~The expected~~ Similarly, a random matrix is a matrix whose elements are random variables. The expected value of a random matrix (vector) is defined as-

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$$

$$E(X) = \begin{bmatrix} E(X_{11}) & E(X_{12}) & \dots & E(X_{1p}) \\ E(X_{21}) & E(X_{22}) & \dots & E(X_{2p}) \\ \vdots & & & \\ E(X_{n1}) & E(X_{n2}) & \dots & E(X_{np}) \end{bmatrix}$$

$$E(X) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_p) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

Let $X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$, $X_1' = [X_1 \; X_2 \; \cdots \; X_p]$ is a $p \times 1$ random vector.

$$E(X) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_p) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} = \mu$$

→ mean vector

For population

Covariance Matrices:- (Dispersion matrix)

For population

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & & & \\ \sigma_{1p} & \sigma_{2p} & \cdots & \sigma_{pp} \end{bmatrix} =$$

$$\Sigma = E[(X-\mu)(X-\mu)^T]$$

μ & Σ as the population mean (vector) & population Var-cov. matrix.

$$= E\begin{bmatrix} (X_1-\mu_1) & (X_1-\mu_1)(X_2-\mu_2) \\ (X_2-\mu_2)(X_1-\mu_1) & (X_2-\mu_2)^2 \end{bmatrix}$$

$$= \begin{bmatrix} E(X_1-\mu_1)^2 & E(X_1-\mu_1)(X_2-\mu_2) \\ E(X_2-\mu_2)(X_1-\mu_1) & E(X_2-\mu_2)^2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

$\sigma_{11} = \text{Var}(X_1)$
$\sigma_{22} = \text{Var}(X_2)$

→ variances of X_1 & X_2

$\text{Cov}(X_1, X_2) = \sigma_{12} = \sigma_{21}$

& the population correlation coefficient ρ_{ik}

$$\rho_{ik} = \frac{\sigma_{ik}}{\sqrt{\sigma_{ii}}\sqrt{\sigma_{kk}}}$$

(σ_{ik} → covariance; $\sqrt{\sigma_{ii}}$, $\sqrt{\sigma_{kk}}$ → S.D)

S.D $(X_i) = \sqrt{\sigma_{ii}}$

It measures the amount of linear association between the random variables X_i & X_k.

σ_{11}

$$V(X_i) = \frac{1}{n}\Sigma(x_i - \bar{x})^2$$

$$V(X_2) = \frac{1}{n}\Sigma(x_2 - \bar{x}_2)^2$$

$$\Sigma = \begin{bmatrix} 9 & 3 \\ 3 & 4 \end{bmatrix}$$

$V(X_1) = 9 \Rightarrow$ S.D $= 3$

$V(X_2) = 4 \Rightarrow$ S.D $= 2$

Multivariate Distributions:-

Normal

First discuss Normal dist^n → then comes into MND.

A generalization of the familiar bell-shaped normal density to several dimensions plays a fundamental role in multivariate analysis.

Normal dist^n in one-dimension- $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \quad \mu, \sigma > 0$$

$-\infty < x < \infty$

But what if we want to model multiple continuous variables together - say height, weight & age of students - which may be correlated?

So we need a dist^n that can capture such multivariate behaviour, especially the relationships (covariances) between the variables.

A random vector $X = [X_1 \; X_2 \; \cdots \; X_p]'$ follows a MND with mean vector μ & covariance matrix (dispersion matrix Σ)

$$\underset{\sim}{X} \sim N_p(\underset{\sim}{\mu}, \Sigma).$$

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & & & \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

↓ cov-matrix of X — Var

Mean vector

$$= \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_p) \end{bmatrix}$$

Density function of MND?

$\left(\frac{x-\mu}{\sigma}\right)^2 = (x-\mu)(\sigma^2)^{-1}(x-\mu)$ in the exponent of Univariate Normal density function.

This can be generalized for a $(p \times 1)$ vector x of observations on several variables as —

$$(x-\mu)' \Sigma^{-1} (x-\mu) \longrightarrow \text{Mahalanobis distance}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

So a p-dim normal density for the random vector $x' = [x_1, x_2 - x_p]$ has the form,

$$x' = [x_1 \quad x_2 \; - - x_p]$$

$$\boxed{f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left[\frac{-1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\right]}$$

$$-\infty < x_i < \infty \; ; \; i = 1, 2, - - p.$$

$$\boxed{x \sim N_p(\mu, \Sigma)} \longrightarrow \text{Notation} \longrightarrow \text{which is analogous to the normal density in univariate case.}$$

↓ Random vector

Properties of MND :-

The following are true for a random vector x having a MND :-

(1) Linear combinations of the components of x are normally distributed.

(2) All subsets of the components of x have a MND.

(3) Zero covariance implies that the corresponding components are independent.

(4) The conditional distributions of the components are multivariate normal.

Result :- If x is distributed as $N_p(\mu, \Sigma)$, then any linear combination of variables $a'x = a_1 x_1 + a_2 x_2 + - - + a_p x_p$ is distributed as $N(a'\mu, a'\Sigma a)$.

Ex: Consider the linear combination $a'x$ of a RV and a random vector X determined by the choice $a' = [1, 0, \ldots 0]$

$$a'x = [1, 0, \ldots 0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = x_1$$

and

$$a'\mu = [1, 0, \ldots 0] \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} = \mu_1$$

also we have

$$a'\Sigma a = [1, 0, \ldots 0] \begin{bmatrix} \sigma_{11} & \sigma_{12} & \ldots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \ldots & \sigma_{2p} \\ \vdots & & & \\ \sigma_{p1} & \sigma_{p2} & \ldots & \sigma_{pp} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \sigma_{11}$$

& it follows the result X_1 is distributed as $N(\mu_1, \sigma_{11})$.

Result:-2 It is for several linear combinations of a multivariate normal vector X.

$$\underset{(q \times p)}{A} \underset{(p \times 1)}{X} = \begin{bmatrix} a_{11}x_1 + \ldots + a_{1p}x_p \\ a_{21}x_1 + \ldots + a_{2p}x_p \\ \vdots \\ a_{q1}x_1 + \ldots + a_{qp}x_p \end{bmatrix}$$

are distributed as $N_q(A\mu, A\Sigma A')$.

Ex:- X is distributed as $N_3(\mu, \Sigma)$. $(p = 3)$

Find the dist^n of $\begin{bmatrix} x_1 - x_2 \\ x_2 - x_3 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad A' = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$A\mu = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \end{bmatrix}$$

$$A\Sigma A' = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

(6)

$$= \begin{bmatrix} \sigma_{11} - 2\sigma_{12} + \sigma_{22} & \sigma_{12} + \sigma_{23} - \sigma_{22} - \sigma_{13} \\ \sigma_{12} + \sigma_{23} - \sigma_{22} - \sigma_{13} & \sigma_{22} - 2\sigma_{23} + \sigma_{33} \end{bmatrix}$$

Result:- (3) All subsets of X are normally distributed. If we respectively partitioned X, its mean vector M & Σ as –

$$X_{(p\times 1)} = \begin{bmatrix} X_1 \\ (q\times 1) \\ X_2 \\ (p-q)\times 1 \end{bmatrix}, \quad M_{(p\times 1)} = \begin{bmatrix} M_1 \\ (q\times 1) \\ M_2 \\ (p-q)\times 1 \end{bmatrix}, \quad \Sigma_{(p\times p)} = \left[\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ (q\times q) & q\times(p-q) \\ \hline \Sigma_{21} & \Sigma_{22} \\ (p-q)\times q & (p-q)\times(p-q) \end{array}\right]$$

then X_1 is distributed as $N_q(M, \Sigma_{11})$

Eg:- $X \sim N_5(M, \Sigma)$. Find the dist^n of $\begin{bmatrix} X_2 \\ X_4 \end{bmatrix}$.

Subset

$$X_1 = \begin{bmatrix} X_2 \\ X_4 \end{bmatrix}, \quad M_1 = \begin{bmatrix} M_2 \\ M_4 \end{bmatrix}, \quad \Sigma_{11} = \begin{bmatrix} \sigma_{22} & \sigma_{24} \\ \sigma_{42} & \sigma_{44} \end{bmatrix}$$

& note that with this assignment X, M & Σ can respectively be rearranged & partitioned as,

$$X = \begin{bmatrix} X_2 \\ X_4 \\ \hline X_1 \\ X_3 \\ X_5 \end{bmatrix} \begin{matrix} \\ \\ \\ \rightarrow X_1 \\ \\ \rightarrow X_2 \end{matrix}, \quad M = \begin{bmatrix} M_2 \\ M_4 \\ \hline M_1 \\ M_3 \\ M_5 \end{bmatrix} \begin{matrix} \rightarrow M_2 \\ \\ \\ \rightarrow M_2 \\ \\ \end{matrix}$$

$$\Sigma = \left[\begin{array}{cc|ccc} \sigma_{22} & \sigma_{24} & \sigma_{21} & \sigma_{23} & \sigma_{25} \\ \sigma_{42} & \sigma_{44} & \sigma_{14} & \sigma_{34} & \sigma_{45} \\ \hline \sigma_{21} & \sigma_{14} & \sigma_{11} & \sigma_{13} & \sigma_{15} \\ \sigma_{23} & \sigma_{34} & \sigma_{13} & \sigma_{33} & \sigma_{35} \\ \sigma_{25} & \sigma_{45} & \sigma_{15} & \sigma_{35} & \sigma_{55} \end{array}\right]$$

Σ_{11} (2×2), Σ_{12} (2×3), Σ_{21} (3×2), Σ_{22} (3×3)

Suppose dist^n of $\begin{bmatrix} X_2 \\ X_3 \end{bmatrix}$

$$N_2\left[\begin{pmatrix} M_2 \\ M_3 \end{pmatrix}, \begin{pmatrix} \Sigma_{22} & \Sigma_{23} \\ \Sigma_{32} & \Sigma_{33} \end{pmatrix}\right]$$

$\begin{bmatrix} X_1 \\ X_5 \end{bmatrix}$

$$N_2\left[\begin{pmatrix} M_1 \\ M_5 \end{pmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{15} \\ \Sigma_{51} & \Sigma_{55} \end{bmatrix}\right]$$

Thus, $X_1 = \begin{bmatrix} X_2 \\ X_4 \end{bmatrix}$, have the distribution

$$N_2(\mu_1, \Sigma_{11}) = N_2\left(\begin{bmatrix} \mu_2 \\ \mu_4 \end{bmatrix}, \begin{bmatrix} \sigma_{22} & \sigma_{24} \\ \sigma_{42} & \sigma_{44} \end{bmatrix}\right)$$

Result:- (a) If X_1 & X_2 are independent, then
$(q_1 \times 1)$ $(q_2 \times 1)$
$\text{cov}(X_1, X_2) = 0$; a $(q_1 \times q_2)$ matrix of zeros.

(b) If $\begin{bmatrix} X_1 \\ \hline X_2 \end{bmatrix}$ is $N_{q_1+q_2}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \left[\begin{array}{c|c} \Sigma_{11} & \Sigma_{12} \\ \hline \Sigma_{21} & \Sigma_{22} \end{array}\right]\right)$ then
X_1 & X_2 are independent iff $\Sigma_{12} = 0$.

(c) If X_1 & X_2 are independent & are distributed as $N_{q_1}(\mu_1, \Sigma_{11})$ & $N_{q_2}(\mu_2, \Sigma_{22})$ resp. then $\begin{bmatrix} X_1 \\ \hline X_2 \end{bmatrix}$ has a MND, $N_{q_1+q_2}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \left[\begin{array}{c|c} \Sigma_{11} & 0 \\ \hline 0 & \Sigma_{22} \end{array}\right]\right)$.

Eg:- $X \sim N_3(\mu, \Sigma)$ with

$$\Sigma = \begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline X_1 & 4 & 1 & 0 \\ X_2 & 1 & 3 & 0 \\ \hline X_3 & 0 & 0 & 2 \end{array}$$

Are X_1 & X_2 independent? What about (X_1, X_2) & X_3?

Sol: Since X_1 & X_2 have covariance $\sigma_{12} = 1 \neq 0$
$\Rightarrow$ They are not independent.

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \hline X_3 \end{bmatrix}, \quad \Sigma = \left[\begin{array}{cc|c} 4 & 1 & 0 \\ 1 & 3 & 0 \\ \hline 0 & 0 & 2 \end{array}\right] = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

we see that $X_1 = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ & X_3 have covariance matrix $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Therefore (X_1, X_2) & X_3 are independent. $\Rightarrow X_3$ is ind. of X_1 & also of X_2.

Numerical problems based on the properties

Q.1 Given $M = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 2 & 1 & 0.5 \\ 1 & 3 & 1 \\ 0.5 & 1 & 2.5 \end{bmatrix}$

$Y = 2X_1 + 3X_2 - X_3 \rightarrow$ (LC)

Find the mean vector & covariance matrix of Y.

Sol: Mean vector $= a'M$, Cov. matrix $= a'\Sigma a$

$a' = [2 \quad 3 \quad -1]$; $a = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$

$a'M = [2 \quad 3 \quad -1] \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix} = 8 - 3 - 5 = 5 - 5 = \boxed{0}$

$a'\Sigma a = [2 \quad 3 \quad -1] \begin{bmatrix} 2 & 1 & 0.5 \\ 1 & 3 & 1 \\ 0.5 & 1 & 2.5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$

$\boxed{a'\Sigma a = 41.5}$ done in the class

Q.2 Let $X \sim N_3(M, \Sigma)$ be a vector where $M = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$\Sigma = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 5 & 0 \\ 1 & 0 & 3 \end{pmatrix}$. Define $Y_1 = 2X_1 + X_2 - X_3$

$Y_2 = -X_1 + 3X_2 + 2X_3$

Sol: Given, $A' = \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ -1 & 2 \end{bmatrix}$, $A^0 = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & 2 \end{bmatrix}$

Mean $= A^0 M$

$\begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

3×2

Cv. matrix $A^0 \Sigma A' = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 & 1 \\ 2 & 5 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ -1 & 2 \end{bmatrix}$

$3 \times 3 \quad 3 \times 2$

$A\Sigma A' = \begin{bmatrix} * & * \\ * & * \end{bmatrix}_{2 \times 3} \quad 3 \times 2$

Ques- Let X be $N_3(\mu, \Sigma)$ with $\mu' = [2 \ -3 \ 1]$ & $\Sigma = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$. Find the distribution of $3X_1 - 2X_2 + X_3$.

Ques- Based on independence :-

(1) $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$, $\Sigma = \begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline X_1 & -1 & -2 & 0 \\ X_2 & -2 & 5 & 0 \\ X_3 & 0 & 0 & 2 \end{array}$

Check the independence of the following random variables & explain.

(i) X_1 & X_3 (ii) X_2 & X_3

(iii) X_1 & (X_2, X_3) (iv) X_2 & (X_1, X_3)

(v) X_3 & (X_1, X_2)

Sol: For independence only Σ matrix is required.

(i) $\Sigma_{13} = 0 \Rightarrow$ ind.

(ii) $\Sigma_{23} = 0 \Rightarrow$ ind.

(iii) X_1 & (X_2, X_3) Cov. matrix is $[-2 \ \ 0]$ Not independent.

(v) X_3 & (X_1, X_2)

$$\begin{array}{c|cc|c} & X_1 & X_2 & X_3 \\ \hline X_1 & -1 & -2 & 0 \\ X_2 & -2 & 5 & 0 \\ X_3 & 0 & 0 & 2 \end{array} = \begin{bmatrix} 0 & 0 \end{bmatrix} \Rightarrow \text{independent}$$

(iv) X_2 & $(X_1, X_3) \Rightarrow [-2 \ \ 0]$ Not independent

OR II Method

$$\begin{array}{c} \\ X_2 \\ X_1 \\ X_3 \end{array} \begin{bmatrix} -2 & 5 & 0 \\ -1 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{\text{Col.}\Rightarrow} \begin{array}{c} \\ X_2 \\ X_1 \\ X_3 \end{array} \begin{array}{c} \begin{array}{cc} X_1 & X_3 \end{array} \\ \begin{bmatrix} 5 & 0 \\ -2 & 0 \\ -1 & 0 \\ 0 & 2 \end{bmatrix} \end{array}$$

$[-2 \ \ 0] \Rightarrow$ Not ind.

ues- Let X be a $N_3(\mu, \Sigma)$ with $\mu' = [-3 \;\; 1 \;\; 4]$ &

$$\Sigma = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \end{array} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left[\begin{array}{ccc} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{array}\right] \end{array}$$

Which of the following random variables are independent? Explain.

(i) X_1 & X_2 (ii) X_2 & X_3 (iii) (X_1, X_2) and X_3

(iv) $\frac{X_1 + X_2}{2}$ and X_3 (v) X_2 & $\left(X_2 - \frac{5}{2}X_1 - X_3\right)$ → Calculate it

Sol: (i) Not ind. (ii) 0 → ind. (iii) $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ → ind.

(iv) $\left(\frac{X_1 + X_2}{2}\right)$ & X_3 → For this Because $\left(\frac{X_1}{2} + \frac{X_2}{2}\right)$ is in the form of a linear combin

$\frac{1}{2}X_1 + \frac{1}{2}X_2 + 0X_3$ $0X_1 + 0X_2 + 1X_3$

we can simplify it using defⁿ of linear combination. So the cov. matrix will be,

$A \Sigma A'$ where $A = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A' = \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \\ 0 & 1 \end{bmatrix}$.

$$A\Sigma A' = \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{matrix} 3\times3 \end{matrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \\ 0 & 1 \end{bmatrix}$$

3×2

$$A\Sigma A' = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 1/2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A\Sigma A' = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 - 1 + 0 & 0 + 0 + 0 \\ -1 + 5/2 + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & 0 + 0 + 2 \end{bmatrix}$$

3×3 3×2

$-2 \times \frac{1}{2}$

$$= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1/2 & 0 \\ 3/2 & 0 \\ 0 & 2 \end{bmatrix}$$

2×3 3×2

$-1 \times \frac{5}{2} = -\frac{2+5}{2}$

$\frac{1}{2} \times \frac{3}{2} \;\; \frac{3}{4} - \frac{1}{4} = \frac{2}{4}$

$$= \begin{bmatrix} -1/4 + 3/4 + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & 0 + 0 + 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \\ 0 \rightarrow \text{ind.} & \end{bmatrix}$$

Mean & Variance of Conditional distribution:-

Take a simple example first to understand the conditional ~~vari~~ distribution

Student	Study Hours (Y)	Marks (X)
A	2	50
B	2	55
C	2	60
D	5	70
E	5	75
F	5	80

We want to know the distribution of marks (X) given the no. of study hours (Y).

Here the condition is on Study Hours Y.

$Y = 2$ hours $\Rightarrow$ Marks $= \{50, 55, 60\}$

Each occurs once out of 3 students.

So prob. of each $= 1/3$

This gives the conditional dist^n of X given $Y = 2$.

Similarly, Condition $Y = 5$ hours.

Marks $= \{70, 75, 80\}$ each with prob. $1/3$.

So the conditional dist^n tells us how marks are distributed once we already know how many hours a student studied.

By this we ~~str~~ restrict our focus to a smaller

but $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ be distributed as $N_p(M, \Sigma)$ with

$M = \begin{bmatrix} M_1 \\ M_2 \end{bmatrix}$ & $\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$. Then the

conditional dist^n of X_1 given that $X_2 = x_2$ is normal & has

$$\text{Mean} = M_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - M_2)$$

and $$\text{Variance} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

Value of Var-Cov does not depend on the value x_2.

Eg-① Let $X \sim N_3(M, \Sigma)$ where $M = [1 \quad -1 \quad 2]'$

& $\Sigma = \begin{bmatrix} 4 & 0 & -1 \\ 0 & 5 & 0 \\ -1 & 0 & 2 \end{bmatrix}$. Find conditional mean & conditional variance of –

(i) x_1, for given $X_2 = x_2$ or $(X_1 \mid X_2 = x_2)$

(ii) $(X_2 \mid X_1 = x_1)$ (iii) $(X_1 \mid X_3 = x_3)$.

Sol: (i) $(X_1 \mid X_2 = x_2)$

$$\text{Mean} = M_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - M_2)$$
$$= 1 + 0(\quad) = 1$$
$$\text{Variance} = 4 - 0(\quad) = 4$$
$$= N(1, 4)$$

(ii) $(X_2 \mid X_1 = x_1)$

$$\text{Mean} = M_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - M_1)$$
$$= -1 + 0(\quad) = -1$$

$$\text{Var} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$
$$= 5 - 0(\quad) = 5$$

$(X_2 \mid X_1 = x_1) \sim N(-1, 5)$

(iii) Try yourself. → Mean $= \frac{4 - x_3}{2}$

Note:- Conditional mean can be negative. but conditional variance can't be negative. it is explained as an expected squared deviation & a square ≥ 0.

↓

conditional mean can take any real value – negative or positive. depending on data. e.g. in the avg. bank balance in a city is zero.

Eg-② $X \sim N_3(M, \Sigma)$ where $M = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 5 & 2 \\ 0 & 2 & 3 \end{bmatrix}$.

Find conditional mean & variance for X_1 given $(X_2 = 2, X_3 = 3)$.

Sol:

$$\Sigma = \begin{array}{c|cc} & x_1 & x_2, x_3 \\ \hline x_1 & 4 & 1 \quad 0 \\ \hline x_2 & 1 & 5 \quad 2 \\ x_3 & 0 & 2 \quad 3 \end{array} \rightarrow \text{Partition of the } \Sigma \text{ matrix}$$

Σ_{11}, Σ_{12}, Σ_{21}, Σ_{22}

$\Sigma_{11} = 4, \quad \Sigma_{12} = 1$

$x_1 = x_1$, $x_2 = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix}$

$$\text{Conditional mean} = M_1 + \Sigma_{12}\Sigma_{22}^{-1}\begin{bmatrix} x_2 - M_2 \\ x_3 - M_3 \end{bmatrix}$$
$$= 1 + \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}^{-1}\begin{bmatrix} 2-2 \\ 3-3 \end{bmatrix}$$
$$= 1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \frac{1}{15-4}\begin{bmatrix} 3 & -2 \\ -2 & 5 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$$

Conditional Variance:-

$\tilde{x}_1 \to x_3$

$\tilde{x}_2 \to \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\Sigma_{12} = \Sigma_{11} - \Sigma_{12}\,\Sigma_{22}^{-1}\,\Sigma_{21}$$

$$= 2\begin{bmatrix} \cdot \end{bmatrix} - \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= 2 - \begin{bmatrix} 1 & 2 \end{bmatrix}\cdot\begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= 2 - \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} 3/2 - 1 \\ -1/2 + 1 \end{bmatrix} = 2 - \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$= 2 - (1/2 + 1)$$

$$= 2 - 3/2 = \frac{4-3}{2} = 1/2 = \underline{0.5}$$

$$(X_3 \mid X_1 = x_1, X_2 = x_2) \sim N\left[(0.5x_1 + 0.5x_2 + 1.5),\ 0.5\right]$$ Ans

Ques- The Var-cov matrix of the random vector $X = [X_1 \ \ X_2 \ \ X_3]'$ is given by

$$\Sigma = \begin{bmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{bmatrix}$$

. Find the correlation matrix & also evaluate the correlation bitwn

X_1 & $\frac{X_2 + X_3}{2}$.

Sol:-

$\sigma_{11} = 25, \ \sigma_{22} = 4, \ \sigma_{33} = 9$ (Variances)

$$\rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\,\sigma_{jj}}}$$

$$\rho_{12} = \frac{\sigma_{12}}{\sqrt{\sigma_{11}\,\sigma_{22}}} = \frac{-2}{\sqrt{25 \times 4}} = \frac{-2}{10}$$

$$R = \begin{bmatrix} 1 & -0.2 & 0.266 \\ -0.2 & 1 & 0.166 \\ 0.266 & 0.166 & 1 \end{bmatrix} \to \text{Correlation matrix}$$

$$\frac{-2}{5 \times 2} = \frac{-2}{10} = \frac{-1}{5} = \underline{-0.2}$$

$$\rho_{23} = \frac{1}{2 \times 3} = 1/6 = 0.166$$

$$\rho_{13} = \frac{\sigma_{13}}{\sqrt{\sigma_{11}}\,\sqrt{\sigma_{33}}} = \frac{4}{5 \times 3} = 4/15 = 0.266$$

correlation bitwn X_1 & $\frac{X_2 + X_3}{2}$.

Correlation (X_1, Y) where $Y = \frac{X_2 + X_3}{2}$ → linear combination

$$\rho(X_1, Y) = \frac{Cov(X_1, Y)}{\sqrt{\sigma_{X_1}}\sqrt{\sigma_Y}} = \frac{Cov(X_1, Y)}{S.D(X_1)\ S.D(Y)}$$

$$Cov\left(X_1, \frac{X_2 + X_3}{2}\right) = ?$$

$$Cov\left(X_1, \frac{X_2 + X_3}{2}\right) = \frac{1}{2}Cov(X_1, X_2) + \frac{1}{2}Cov(X_1, X_3)$$

$$= \frac{1}{2}\sigma_{12} + \frac{1}{2}\sigma_{13}$$

$$= \frac{1}{2} \times (-2) + \frac{1}{2} \times 4 = -1 + 2 = 1$$

$$\rho(X_1, Y) = \frac{1}{\sqrt{25}\sqrt{\sigma_{YY}}}$$

$$Var\left(\frac{X_2 + X_3}{2}\right) = \frac{1}{4}\left[V(X_2) + V(X_3) + 2Cov(X_2, X_3)\right]$$

$$= \frac{1}{4}\left[4 + 9 + 2 \times 1\right] = \frac{1}{4}(15) = \frac{15}{4} = 3.75$$

$$\rho(X_1, Y) = \frac{1}{5 \times \sqrt{3.75}} = \frac{1}{5 \times 1.936} = \frac{1}{9.68} = 0.1033$$

Positive correlation

II method

$$X_1 \longrightarrow 1X_1 + 0X_2 + 0X_3$$
$$Y = 0X_1 + \frac{1}{2}X_2 + \frac{1}{2}X_3$$

Use this method in class

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}, \quad A' = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{bmatrix} \quad (3 \times 2)$$

Cov matrix $A\Sigma A' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 25+0+0 & 0-1+2 \\ -2+0+0 & 0+2+1/2 \\ 4+0+0 & 0+1/2+9/2 \end{bmatrix} \quad (3\times3)\ (3\times2)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 25 & 1 \\ -2 & 5/2 \\ 4 & 5 \end{bmatrix} \quad (2\times3)\ (3\times2)$$

$\frac{5}{2} \times \frac{1}{2}$ $\quad \frac{10}{2}$ $\quad \Sigma_{11}$ $\quad \Sigma_{12}$

$$= \begin{bmatrix} 25+0+0 & 1+0+0 \\ 0-1+2 & 0+\frac{5}{4}+\frac{5}{2} \end{bmatrix} = \begin{matrix} X_1 \\ Y \end{matrix}\begin{bmatrix} 25 & 1 \\ 1 & 15/4 \end{bmatrix}$$

Y_1 $\quad$ LC

1 → 3.75

Σ_{21} $\quad \Sigma_{22}$

$\frac{5+10}{4}$

Now $\text{Cov}(X_1, Y) = \sigma_{12} = 1$

$$\rho(X_1, Y) = \frac{\sigma_{12}}{\sqrt{\sigma_{11} \times \sigma_{22}}} = \frac{1}{\sqrt{25} \times \sqrt{3.75}} = \frac{1}{5 \times 1.936}$$

$$= 0.1033 \quad \text{Ans}$$

Eg (5) Conditional dist^n :-

Let $X \sim N_3(M, \Sigma)$ with $M = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$, $\Sigma = \begin{array}{c} \begin{matrix} X_1 & X_2 & X_3 \end{matrix} \\ \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{array}$.

Find the conditional dist^n of X_2 given $X_1 = x_1$ & $X_3 = x_3$.

Sol:-

$$\begin{array}{c} \begin{matrix} X_1 & X_2 & X_3 \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{array}$$

First Rearrange

$$\begin{bmatrix} -2 & 5 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{array}{c} \begin{matrix} \Sigma_{11} & X_2 & X_1 & X_3 \end{matrix} \\ \text{Cov} \begin{matrix} X_2 \\ X_1 \\ X_3 \end{matrix} \left[\begin{array}{c|cc} 5 & -2 & 0 \\ \hline -2 & 1 & 0 \\ 0 & 0 & 2 \end{array}\right] \\ \begin{matrix} \Sigma_{21} & & \Sigma_{22} \end{matrix} \end{array} \quad \Sigma_{12}$$

New Σ matrix for calculation

(i) Conditional mean

$$\tilde{M}_1 + \tilde{\Sigma}_{12}\, \tilde{\Sigma}_{22}^{-1} (\tilde{x}_2 - \tilde{M}_2)$$

$$1 + \begin{bmatrix} -2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} x_1 - (-3) \\ x_3 - 4 \end{bmatrix}$$

$$= 1 + \begin{bmatrix} -2 & 0 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 + 3 \\ x_3 - 4 \end{bmatrix}$$

$\tilde{x}_1 \rightarrow x_2$

$\tilde{x}_2 \Rightarrow \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$

$\tilde{M}_1 \rightarrow 1$

$\tilde{M}_2 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$

$\frac{1}{2}$

Calculate it & Try Variance also.

Hotelling's T^2 Statistic

In Univariate Case:- $\mu \rightarrow$ Population mean

& $\mu_0 \rightarrow$ some specific value of μ.

Then we test the hypothesis:

$H_0 : \mu = \mu_0$ (specific value)

Null hypothesis

& $H_1 : \mu \neq \mu_0$ (Alternative Hypo.)

For small samples (say ≤ 30) we use t-distribution

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

let $x_1, x_2 \ldots x_n$ are random samples.

To test the single mean value

where $\bar{x} \rightarrow$ Sample mean

$\mu \rightarrow$ Population mean

$s \rightarrow$ sample standard deviation

Now, looking into multivariate case,

$\bar{x} \rightarrow \bar{X}$, $\mu \rightarrow \mu$, $s^2 = S$, $n \rightarrow n$

$S \rightarrow$ Sample Standard

Now, $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \sqrt{n}\,(\bar{x} - \mu)\,s^{-1}$

If we square $t^2 = n(\bar{x} - \mu)^2 (s^2)^{-1}$

$= n(\bar{x} - \mu)(s^2)^{-1}(\bar{x} - \mu) = n(\bar{x} - \mu)^2 (s^2)^{-1}$

$$T^2 = n(\bar{X}-M)'(S)^{-1}(\bar{X}-M)$$

or $T^2 = n(\bar{X}-M)^T (S)^{-1}(\bar{X}-M) \sim \frac{(n-1)p}{n-p} F_{p,n-p}$

$\forall\ n-p < 30$.

This is Hotelling's T^2 statistic

$$\boxed{T^2 = n(\bar{X}-M)'(S)^{-1}(\bar{X}-M)}$$

where $\bar{X} \sim N_p(M, \Sigma/n)$

& $S \sim W_p(n-1, \Sigma)$

For $M = M_0$

$$T^2 = n(\bar{X}-M_0)'S^{-1}(\bar{X}-M_0)$$

Eg-① A random sample with $n = 20$ were collected from a bivariate normal process. The sample mean vector $\bar{X}$ & cov. matrix S is given as follows:

$$\bar{X} = \begin{bmatrix} 10 \\ 20 \end{bmatrix}, \quad S = \begin{bmatrix} 40 & -50 \\ -50 & 100 \end{bmatrix}$$

(i) Obtain Hotelling's T^2.

(ii) What will be the distribution of it?

Sol: We know that,

$$T^2 = n(\bar{X}-M)^T S^{-1}(\bar{X}-M)$$

$$\bar{X} = \begin{bmatrix} 10 \\ 20 \end{bmatrix}, \quad S = \begin{bmatrix} 40 & -50 \\ -50 & 100 \end{bmatrix}$$

$$\bar{X}-M = \begin{bmatrix} 10 \\ 20 \end{bmatrix} - \begin{bmatrix} M_1 \\ M_2 \end{bmatrix} = \begin{bmatrix} 10-M_1 \\ 20-M_2 \end{bmatrix}$$

$$S^{-1} = \frac{1}{|S|}\text{adj}(S) = \frac{1}{1500}\begin{bmatrix} 100 & 50 \\ 50 & 40 \end{bmatrix} = \begin{bmatrix} 1/15 & 1/30 \\ 1/30 & 2/75 \end{bmatrix}$$

$$\therefore T^2 = 20\begin{bmatrix} 10-M_1 & 20-M_2 \end{bmatrix}\begin{bmatrix} 1/15 & 1/30 \\ 1/30 & 2/75 \end{bmatrix}\begin{bmatrix} 10-M_1 \\ 20-M_2 \end{bmatrix}$$

Ans $T^2 = 20\begin{bmatrix} 10-M_1 & 20-M_2 \end{bmatrix}\begin{bmatrix} \frac{1}{15}(10-M_1) + \frac{1}{30}(20-M_2) \\ \frac{1}{30}(10-M_1) + \frac{2}{75}(20-M_2) \end{bmatrix}$

(2)

$$T^2 = 20\begin{bmatrix}10-\mu_1 & 20-\mu_2\end{bmatrix}\begin{bmatrix}\frac{2}{3}-\frac{1}{15}\mu_1+\frac{2}{3}\cdot-\frac{1}{30}\mu_2 \\ \frac{1}{3}-\frac{1}{30}\mu_1+\frac{8}{15}-\frac{2}{75}\mu_2\end{bmatrix}$$

$$= 20\begin{bmatrix}10-\mu_1 & 20-\mu_2\end{bmatrix}_{(1\times 2)}\begin{bmatrix}\frac{1}{15}\times(10-\mu_1)+\frac{1}{30}(20-\mu_2) \\ \frac{1}{30}(10-\mu_1)+\frac{2}{75}(20-\mu_2)\end{bmatrix}_{(2\times 1)}$$

$$= 20\left[\frac{1}{15}(10-\mu_1)^2+\frac{1}{30}(10-\mu_1)(20-\mu_2)+(20-\mu_2)\times\frac{1}{30}(10-\mu_1)+(20-\mu_2)^2\times\frac{2}{75}\right]$$

$$= 1.33(10-\mu_1)^2+\frac{2\times 20}{30}(10-\mu_1)(20-\mu_2)+\frac{2\times 20}{75}(20-\mu_2)^2$$

$$\boxed{T^2 = 1.33(10-\mu_1)^2+(1.33)(10-\mu_1)(20-\mu_2)+(0.53)(20-\mu_2)^2}$$

(ii) $T^2 \sim \frac{(n-1)p}{n-p}F_{n,n-p}$

$$= \frac{(20-1)\times 2}{20-2}F_{2,20-2}$$

$$= \frac{19\times 2}{18}F_{2,18} = \frac{19}{9}(F_{2,18}) \rightarrow \text{distribution}$$

$$T^2 = \frac{19}{9}\times 3.56 = 7.515 \qquad \frac{19}{9}\times 3.56 = 7.51$$

This value is called critical value ~~or tabulated val~~ of T^2 statistic.

$T^2 \rightarrow$

Ques- Let the data matrix for a random sample of size $n=3$ from bivariate normal dist^n be $X = \begin{bmatrix}6 & 9\\ 10 & 6\\ 8 & 3\end{bmatrix}$. Evaluate the observed T^2 for $\mu_0' = [9 \quad 5]$

or $\mu_0 = \begin{bmatrix}9\\5\end{bmatrix}$

What is the sampling dist^n of T^2 in this case?

Example:- X_1, X_2, X_3 Three components

$X_1 \rightarrow$ Sweat Rate, $X_2 =$ Sodium content

$X_3 \rightarrow$ Potassium content were measured.

Perspiration from 20 healthy Females was analyzed.

Test the Hypothesis $H_0: M_0' = [4 \quad 50 \quad 10]$

against $H_1: M_0' \neq [4 \quad 50 \quad 10]$

at level of significance $\alpha = 0.05$. (5%).

Given that

$$\bar{X} = \begin{bmatrix} 4.64 \\ 45.4 \\ 9.965 \end{bmatrix}$$

Based information on some real data set

$$S = \begin{bmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{bmatrix}$$

& $S^{-1} = \begin{bmatrix} 0.586 & -0.022 & 0.258 \\ -0.022 & 0.006 & -0.002 \\ 0.258 & -0.002 & 0.402 \end{bmatrix}$

We evaluate,

$$T^2 = 20 [4.640 - 4 \quad 45.4 - 50 \quad 9.965 - 10]$$

$$\begin{bmatrix} 0.586 & -0.022 & 0.258 \\ -0.022 & 0.006 & -0.002 \\ 0.258 & -0.002 & 0.402 \end{bmatrix} \begin{bmatrix} 4.640 - 4 \\ 45.4 - 50 \\ 9.965 - 10 \end{bmatrix}$$

$$= 20 [0.640 \quad -4.6 \quad -0.035] \begin{bmatrix} 0.467 \\ -0.042 \\ 0.160 \end{bmatrix}$$

$$\boxed{T^2 = 9.74}$$

This is the observed value of T^2.

$T^2 = 9.74$

The dist^n of T^2 is $\frac{(n-1)p}{n-p} F_{p,n-p}$

$\frac{(20-1) \times 3}{20-3} = \frac{19 \times 3}{17} F_{3,17}$ at 5% lm

$F_{3,17}(0.05)$

$= \frac{57}{17} F_{(3,17)}(0.05)$

$(n-p) < 40$

$(20-3) = 17 < 40$

$= 3.3529 \times 3.197$

$= 10.719$

$T^2_{cal} = 9.74$, T^2 critical value $= 10.719$

$T^2_{cal} > T^2_{critical}$ we reject H_0 at 5% level of Significance.

It means that one or more of the component means or some combinations of means differs too much from the hypothesized value [4 50 10].

Two Sample Hotelling's T^2 Statistic :-

$H_0: M_1 = M_2$ V/S $H_1: M_1 \neq M_2$

$$S_{pooled} = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$$

For this, Hotelling's T^2 statistic,

$$T^2 = (\bar{X}_1 - \bar{X}_2)' \left[\left(\frac{1}{n_1}+\frac{1}{n_2}\right) S_p\right]^{-1} (\bar{X}_1 - \bar{X}_2)$$

$$\boxed{T^2 = \frac{n_1 n_2}{n_1+n_2} (\bar{X}_1 - \bar{X}_2)' S_p^{-1} (\bar{X}_1 - \bar{X}_2)} \sim \frac{(n_1+n_2-2)p}{(n_1+n_2-p-1)} F_{p,}$$

$(n_1+n_2-p-$

$n_1+n_2-p-1 < 30$ ~~$n-p \leq 30$~~ ~~F dist^n~~

$n_1+n_2-p-1 > 30$ ~~$n-p \geq 30$~~ $\Rightarrow \chi^2_p(\alpha)$ (large Sample)

Ques - A Study is conducted to compare the performance of two manufacturing machines. (Machine A & B) on two quality character X_1 & X_2.

The following sample statiscs are obtained from two samples each of size 4.

$$\bar{X}_1 = \begin{bmatrix} 7.5 \\ 5.75 \end{bmatrix}, \quad \bar{X}_2 = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \quad \& \quad S_1 = \begin{bmatrix} 1.67 & 0.67 \\ 0.67 & 0.92 \end{bmatrix}, S_2 = \begin{bmatrix} 0.67 & 0 \\ 0.33 & 0.2 \end{bmatrix}$$

Test at 5% LOS whether there is a significant diff between the mean vectors of the two machines using Hotelling's T^2 statistic at 5% LOS.

Sol: First compute, $S_{pooled} = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$

$$S_p = \frac{3S_1 + 3S_2}{8-2} = \frac{3(S_1+S_2)}{6}$$

$$= \frac{1}{2}[S_1 + S_2] = \frac{1}{2}\begin{bmatrix} 1.17 & 0.5 \\ 0.5 & 0.585 \end{bmatrix} = S_p$$

$$\bar{X}_1 - \bar{X}_2 = \begin{bmatrix} 2.5 \\ 2.75 \end{bmatrix}$$

$$S_p^{-1} = \frac{1}{0.68445 - 0.25}\begin{bmatrix} 0.585 & -0.5 \\ -0.5 & 1.17 \end{bmatrix}$$

$$= \frac{1}{0.43445}\begin{bmatrix} \text{"} \end{bmatrix} = \begin{bmatrix} 1.346 & -1.1508 \\ -1.1508 & 2.694 \end{bmatrix}$$

$$T^2 = \frac{n_1 n_2}{n_1+n_2}(\bar{X}_1 - \bar{X}_2)' S_p^{-1} (\bar{X}_1 - \bar{X}_2)$$

$$= \frac{4\times 4}{4+4}\begin{bmatrix} 2.5 & 2.75 \end{bmatrix}\begin{bmatrix} \quad \end{bmatrix}\begin{bmatrix} 2.5 \\ 2.75 \end{bmatrix}$$

$$= 2\left[\frac{12.97}{}\right] = 25.94$$

$$\boxed{T^2 = 25.94}$$

$T^2_{cal} > T_{tab}$

$\Rightarrow$ Reject H_0 at 5% LOS

$$T^2 \sim \frac{(n_1+n_2-2)p}{(n_1+n_2-p-1)} F_{p,\, n_1+n_2-p-1}$$

$$\sim \frac{(4+4-2)\times 2}{(4+4-2-1)} F_{2,\, 4+4-2-1}$$

$$\boxed{T^2 \sim \frac{12}{5} F_{2,5}(0.05)}$$

$4+4-2-1 = 8-2-1 = 6-1 = 5 < 30$

Now Calculate

$= \frac{12}{5} \times 5.786$

$= 13.89$

$n = n_1 + n_2$

$n = 4+4$

$= 8 < 30$

So F dist

$n_1 + n_2 - p - 1 > 30$

Chi Square

$n_1 + n_2 - p - 1 < 30$

↓

F distⁿ

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Large Sample $\sim \chi^2_p$.

Q. Fifty bars of shop are manufactured in each of two ways. Two chemtrics X_1 = lather & X_2 = mildness are measured. The summary statistic for bars produced by methods 1 & 2 are —

$$\bar{X}_1 = \begin{bmatrix} 8.3 \\ 4.1 \end{bmatrix}, \quad S_1 = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}, \quad \bar{X}_2 = \begin{bmatrix} 10.2 \\ 3.9 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

Test the null hypothesis $H_1 - H_2 = 0$ using Hotelling's T^2 at 5% LOS.

Sol: $n_1 = 50, \quad n_2 = 50$ (Chi square)

$$S_p = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}, \quad S_p^{-1} = \begin{bmatrix} 5/9 & -1/9 \\ -1/9 & 2/9 \end{bmatrix}$$

$$T^2_{cal} = 25 \begin{bmatrix} -1.9 \\ 0.2 \end{bmatrix}' \begin{bmatrix} 5/9 & -1/9 \\ -1/9 & 2/9 \end{bmatrix} \begin{bmatrix} -1.9 \\ 0.2 \end{bmatrix}.$$

$= 25 \times 2.0973$

$\boxed{T^2_{cal} = 52.43} > \chi^2_p(0.05)$ ⇒ Reject H_0. at 5%.

~~$N = n_1 + n_2 - 2 = 100 - 2 = 98 > 30$ large~~

$n_1 + n_2 - p - 1 = 50 + 50 - 2 - 1$

$= 100 - 3 = 97 > 30$

Chi Square

$T^2 \sim \chi^2_p$

$T^2 \sim \chi^2_2(0.05) = 5.99$